

Autonomous Payload Delivery on Vision-Detected Landing Pads

Onboard YOLO Detection for Indoor FPV Drone Delivery

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Docs



Code

01 Introduction

This project extends a ready-built FPV drone with an autopilot function and an onboard AI camera that detects landing pads in order to release a payload over them. This is a step towards delivery drones that identify a designated landing spot autonomously, indoors as well as outdoors, where GPS alone is neither accurate nor available enough.

02 Objectives

- Detect landing pads reliably from a moving drone using onboard vision
- Integrate autopilot and indoor position hold without GPS
- Release the payload via a servo-based drop mechanism

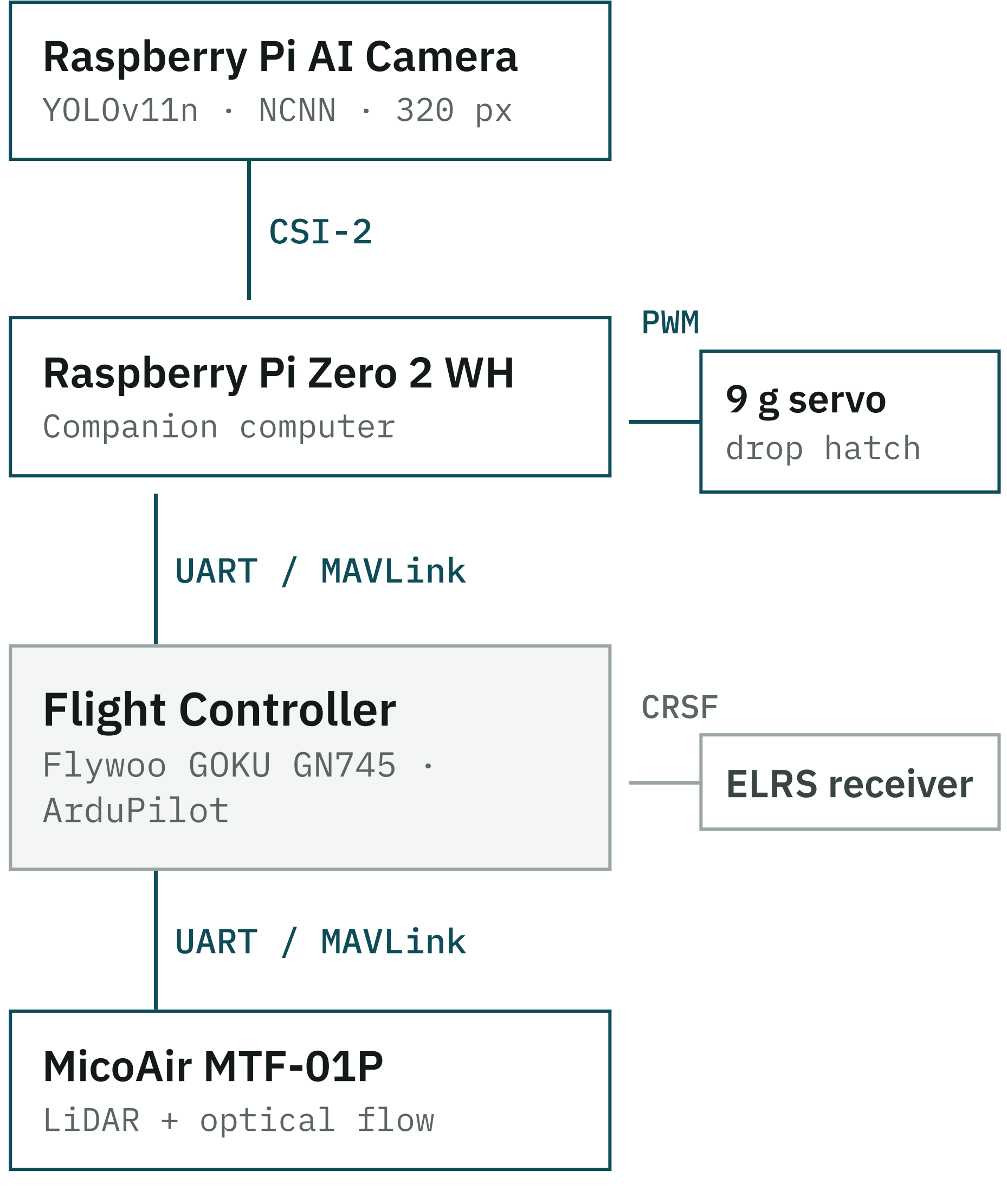
03 System Components

Platform	SpeedyBee BEE35 Pro 3.5" CineWhoop
Flight Controller	Flywoo GOKU GN745, ArduPilot
Companion Computer	Raspberry Pi Zero 2 WH
Sensors	Raspberry Pi AI Camera, MicoAir MTF-01P (LiDAR + optical flow)
Payload	9 g micro servo, 3D-printed frame extension
GPS	HGLRC M100 with compass
Motor	Emax Eco II 2004 3-6S 3000KV
Rotors	Gemfan 90mm D90-5 3.5" Ducted 5-Blade Rotors

IMAGE PLACEHOLDER

Hardware photo: drone with Pi Zero 2 WH, AI camera and servo drop mechanism

04 System Architecture



☐ Added by the team

☐ Pre-existing hardware

05 Mission Sequence

- 1 Command**
The ground station sends the mission command over the ELRS link.
- 2 Search pattern**
The drone flies a preset pattern at hold altitude while the camera streams to the companion computer.
- 3 Detection**
YOLOv11n detects a landing pad. A person in frame blocks the release.
- 4 Approach**
The drone steers towards the detected pad and centers above it.
- 5 Release**
The servo opens the hatch and the payload drops.

ON FAULT

Lost link, no detection within the pattern or low battery: the autopilot returns to launch and lands.

ex. 1

ex. 2

ex. 3

Onboard detections with bounding boxes

06 Autopilot & Position Hold

Migration from Betaflight to ArduPilot enables indoor position and altitude hold without GPS. Optical flow and LiDAR rangefinding from the MicoAir MTF-01P feed the flight controller over UART; sensor configuration and calibration follow. [Stand ergänzen]

07 Challenges

- Limited compute on the Raspberry Pi Zero 2 W: only a few frames per second
- Small dataset, restricted lighting and viewing-angle variation
- Vibration and motion blur during flight
- Indoor magnetic interference from building rebar caused EKF compass yaw resets, disrupting GPS-denied position hold

08 Conclusion

The drone detects landing pads onboard and releases its payload without relying on GPS. One two-class model proved preferable to two parallel models, since the companion computer cannot carry both. Detection holds across changing light and viewing angles; inference speed is what still limits a fully closed-loop approach.

IMAGE PLACEHOLDER

Flight test: indoor position hold over the landing pad